

A Forbidden Composite-Cosine Parameter Is Nevertheless Injective

Counterexample packet for arXiv:math/0501020

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Status: candidate full counterexample, likely valid; human review requested.
For $(n, m) = (3, 2)$, the composite cosine transform with $\lambda = (\alpha, 0)$ is injective on $L^b(V_{3,2})$ exactly when $\alpha \notin 2\mathbb{N}_0$. In particular, $T^{(1,0)}$ is injective even though both entries violate the condition that Ournycheva and Rubin ask might be necessary when $2m > n$. Thus the answer to their necessity question is no.

1 The source question

For $n > m$, Ournycheva and Rubin [1] define

$$(T^\lambda f)(u) = \int_{V_{n,m}} f(v) (u'vv'u)^\lambda dv, \quad u \in V_{n,m}, \quad (1)$$

on right- $O(m)$ -invariant integrable functions. If r is a positive $m \times m$ matrix and $\Delta_j(r)$ is its leading principal $j \times j$ minor, its composite power is

$$r^\lambda = \prod_{j=1}^m \left(\frac{\Delta_j(r)}{\Delta_{j-1}(r)} \right)^{\lambda_j/2}, \quad \Delta_0 = 1. \quad (2)$$

In the convergence domain $\operatorname{Re} \lambda_j > j - m - 1$, their Theorem 5.1 gives injectivity under

$$\lambda_j + m - j \notin \{0, 2, 4, \dots\}, \quad j = 1, \dots, m. \quad (3)$$

They prove (3) necessary for $2m \leq n$ and ask whether it is necessary when $2m > n$.

which follows from (4.17) and (4.15). Suppose that $(T^\lambda f)(v) = 0$ almost everywhere on $V_{n,m}$ for some $\lambda \in \mathfrak{L}$. Then $(T^\lambda f)(y)$, $y = vr^{1/2} \in \mathfrak{M}_{n,m}$, is zero for almost all $y \in \mathfrak{M}_{n,m}$, and (5.8) yields $A(\lambda)(r_*^{-\lambda_* - n_0} f(v), \phi) = 0$. The assumption (5.1) together with $\lambda \in \mathfrak{L}$ imply

$$\operatorname{Re} \lambda_j > j - m - 1 \quad \text{and} \quad \lambda_j \neq j - m, j - m + 2, \dots.$$

Hence λ is not a pole of $\Gamma_\Omega(-\lambda_*)$, and therefore $A(\lambda) \neq 0$. This gives

$$(5.9) \quad (r_*^{-\lambda_* - n_0} f(v), \phi) = 0,$$

the left-hand side being understood in the sense of analytic continuation. By choosing ϕ as in the proof of Corollary 5.4 we obtain $f(v) = 0$ a.e. on $V_{n,m}$.

Let us prove the second statement of the theorem. We first note that for $2m < n$, H -polynomials of every degree k do exist. Hence we can proceed as follows. We observe that the function $\Gamma_\Omega(\mathbf{k}_0 - \lambda_*)$ in the expression of $\mu_k(\lambda)$ has the form

$$\begin{aligned} \Gamma_\Omega(\mathbf{k}_0 - \lambda_*) &= \pi^{m(m-1)/4} \prod_{j=1}^m \Gamma\left(\frac{k - (\lambda_*)_j - j + 1}{2}\right) \\ &= \pi^{m(m-1)/4} \prod_{j=1}^m \Gamma\left(\frac{k + j - \lambda_j - m}{2}\right) \end{aligned}$$

(we remind that $(\lambda_*)_j = \operatorname{Re} \lambda_{m-j+1}$). It has no poles in $\mathfrak{L}$ if

$$(5.10) \quad k > \lambda_j + m - j \quad \text{for all} \quad j = 1, 2, \dots, m.$$

Since $\Gamma_\Omega(\lambda + \mathbf{m}_0)$ also has no poles in $\mathfrak{L}$, then, by (5.5), $T^\lambda P_k = 0$ for all k satisfying (5.10), provided that λ is a pole of $\Gamma_\Omega(-\lambda_*)$ (i.e., the condition (5.1) fails). Thus T^λ is non-injective in this case. $\square$

We do not know if the condition (5.1) is necessary for injectivity of T^λ in the case $2m > n$. We suspect that to answer this question one has to explore T^λ on polynomial representations of the group $SO(n)$ parameterized by highest weights $(m_1, m_2, \dots, m_{\lfloor n/2 \rfloor})$, more general than just $(k, k, \dots, k)$ corresponding to $\mathcal{H}_k$ -spaces; cf. [Str], [TT]. However, if $\lambda_1 = \dots = \lambda_m = \lambda$ then for the λ -cosine transform

$$(T^\lambda f)(u) = \int_{V_{n,m}} f(v) |\det(v'u)|^\lambda dv \quad \left(\text{or} \quad (T^\lambda f)(\xi) = \int_{G_{n,m}} f(\eta) |\eta[\xi]^\lambda d\eta \right),$$

we can give a complete answer.

THEOREM 5.7. *Let $n > m$ and $\operatorname{Re} \lambda > -1$. Then T^λ is injective on $L^b(V_{n,m})$ (or on $L^1(G_{n,m})$) if and only if $\lambda \neq 0, 1, 2, \dots$.*

Figure 1: The source's question and surrounding conditions (PDF page 19).

2 Idea of the proof

Use the smallest over-rank Grassmannian, $G_{3,2}$. A plane is determined by a unit normal w , modulo sign. On the parameter slice $\lambda = (\alpha, 0)$, the first principal minor of $u'vv'u$ is $1 - (w \cdot u_1)^2$, while the second exponent is zero. The allegedly more complicated composite transform therefore becomes a zonal convolution on Sph^2 .

The degree- $2k$ multiplier of this convolution is an elementary Legendre integral. A beta integral and Chu–Vandermonde evaluate it as a product whose only possible zero is $(-\alpha/2)_k$. Hence every even harmonic survives unless α is a nonnegative even integer. Choosing $\alpha = 1$ makes all multipliers nonzero while making both exclusions in (3) fail.

3 The exact over-rank slice

We use $(a)_k = a(a+1)\cdots(a+k-1)$ for the rising factorial.

Lemma 1 (Legendre multiplier). *If $\operatorname{Re} a > -1$ and $k \in \mathbb{N}_0$, then*

$$\int_{-1}^1 (1-t^2)^a P_{2k}(t) dt = B\left(\frac{1}{2}, a+1\right) \frac{(-a)_k (1/2)_k}{k! (a+3/2)_k}. \quad (4)$$

Proof. Put $s = t^2$ and use the terminating representation

$$P_{2k}(\sqrt{s}) = (-1)^k \frac{(1/2)_k}{k!} {}_2F_1\left(-k, k + \frac{1}{2}; \frac{1}{2}; s\right).$$

Termwise beta integration is legitimate because the series is a polynomial. The beta ratio cancels the denominator parameter $1/2$, leaving

$$(-1)^k \frac{(1/2)_k}{k!} B\left(\frac{1}{2}, a+1\right) {}_2F_1\left(-k, k + \frac{1}{2}; a + \frac{3}{2}; 1\right).$$

Chu–Vandermonde turns the last factor into $(a+1-k)_k / (a+3/2)_k = (-1)^k (-a)_k / (a+3/2)_k$, proving (4). $\square$

Theorem 2 (Complete criterion on a forbidden slice). *Let $(n, m) = (3, 2)$, let $\lambda = (\alpha, 0)$, and assume $\operatorname{Re} \alpha > -2$. Then*

$$T^{(\alpha, 0)} : L^b(V_{3,2}) \longrightarrow L^1(V_{3,2})$$

is injective if and only if $\alpha \notin \{0, 2, 4, \dots\}$.

Proof. Identify a plane $\eta \in G_{3,2}$ with a unit normal $w \in \mathbb{S}^2$, where w and $-w$ represent the same plane. Thus $L^b(V_{3,2})$ identifies, up to harmless normalization of measure, with the even functions $F \in L^1(\mathbb{S}^2)$.

Write $u = (u_1, u_2) \in V_{3,2}$. If the columns of v span $w^\perp$, then $vv' = I - ww'$, and hence

$$\Delta_1(u'vv'u) = u'_1(I - ww')u_1 = 1 - (w \cdot u_1)^2. \quad (5)$$

By (2), the second exponent vanishes when $\lambda = (\alpha, 0)$. Consequently

$$(T^{(\alpha, 0)}F)(u) = c \int_{\mathbb{S}^2} F(w) (1 - (w \cdot u_1)^2)^{\alpha/2} d\sigma(w), \quad (6)$$

where $c > 0$ depends only on the measure normalization. The convergence assumption is exactly $\operatorname{Re} \alpha > -2$.

Set $a = \alpha/2$. By the Funk–Hecke formula, the operator in (6) acts on every spherical harmonic of degree $2k$ by a nonzero common normalizing constant times the integral in (4). In the half-plane $\operatorname{Re} a > -1$, the beta factor and all denominator factors in (4) are finite and nonzero. Thus the degree- $2k$ multiplier vanishes exactly when $(-a)_k = 0$.

If $a \notin \mathbb{N}_0$, no even multiplier vanishes. If (6) is zero almost everywhere, Fubini and Funk–Hecke show that every even spherical-harmonic coefficient of F is zero. Since spherical harmonics are uniformly dense in continuous functions on $\mathbb{S}^2$, and F is even, the finite measure $F d\sigma$ is zero; hence $F = 0$ almost everywhere. The transform is injective.

If $a = q \in \mathbb{N}_0$, the kernel $(1-t^2)^q$ is a polynomial of degree $2q$. Every nonzero spherical harmonic of even degree greater than $2q$ is therefore annihilated, so the transform is not injective. $\square$

4 Counterexample to the proposed necessity

Take

$$n = 3, \quad m = 2, \quad \lambda = (1, 0). \quad (7)$$

This lies in the source’s convergence domain because $1 > -2$ and $0 > -1$. But the two quantities excluded by (3) are

$$\lambda_1 + m - 1 = 2, \quad \lambda_2 + m - 2 = 0, \quad (8)$$

so (3) fails in both coordinates. Theorem 2, with $\alpha = 1 \notin 2\mathbb{N}_0$, nevertheless proves that $T^{(1,0)}$ is injective. Therefore condition (3) is not necessary when $2m > n$.

5 Verification, scope, and novelty

The argument answers the source’s yes/no necessity question completely and also determines injectivity on the whole slice $(3, 2, (\alpha, 0))$. It does not classify all multiparameters for every pair $2m > n$; after the negative answer, that sharper classification remains a separate problem.

The accompanying script numerically integrates (4) at $a = 1/2$ for $k = 0, \dots, 8$ and compares with the closed formula at 70-digit precision. This checks transcription and signs but is not used as proof. Expert review should focus on the identification (5)–(6), the multiplier evaluation (4), and the L^1 completeness step.

On 12 August 2026, the four run indexes and bounded exact/current searches for the source identifier, the exact quoted question, ‘composite cosine transform’, ‘ $2m > n$ ’, the authors, and polynomial-representation follow-up found the source and its 2006 restatement, but no later explicit solution of the multiparameter necessity question. Novelty confidence is moderate-high.

Human-review recommendation. Review as a full counterexample. The witness is explicit, lies strictly in the source’s convergence domain, violates both proposed exclusions, and its injectivity follows from a complete elementary multiplier calculation.

References

- [1] E. Ournycheva and B. Rubin, *The Composite Cosine Transform on the Stiefel Manifold and Generalized Zeta Integrals*, arXiv:math/0501020v2 (2005), Theorem 5.1 and the question on PDF page 19.